

Paper Summary

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1 Density estimation and regression analysis on hyperspheres in the presence of measurement error

1.1 The problem

Most of the previous works on analyzing hyperspherical data did not cover a measurement error problem. Even though some works covered measurement errors, these works only covered special hyperspheres. Therefore, this study covers deconvolution density estimation and regression analysis on general $\mathbb{S}^d$, which have not been covered in the literature.

1.2 The setup

In this study, we do not observe X but we only observe $Z = UX$, where U is an unobservable measurement error taking values in $\text{SO}(d+1)$. We want to estimate (i) density f_X based on i.i.d. observations Z_i and (ii) regression function $m : \mathbb{S}^d \rightarrow \mathbb{R}$ such that $Y = m(X) + \epsilon$ based on i.i.d. observations $(Y_i, Z_i) : 1 \leq i \leq n$, where Y is real-valued and ϵ is error term satisfying $E(\epsilon | X) = 0$. Additionally, $U \perp (X, \epsilon)$.

1.3 The main idea

The key idea behind the proposed density estimator is to view the measurement error problem on $\mathbb{S}^d$ as a deconvolution problem using harmonic analysis. Since the spherical harmonics $\{B_q^l : l \in \mathbb{N}_0, 1 \leq q \leq N(d, l)\}$ form an orthonormal basis of $L^2(\mathbb{S}^d)$, the target density can be expanded as

$$f_X = \sum_{l=0}^{\infty} \sum_{q=1}^{N(d,l)} \phi_q^l(f_X) B_q^l.$$

Under the measurement error model $Z = UX$, f_Z satisfies the convolution relation $f_Z = f_U * f_X$, so, at each harmonic degree l ,

$$\phi^l(f_Z) = \tilde{\phi}^l(f_U) \phi^l(f_X).$$

Therefore, when the measurement error distribution f_U is known and $\tilde{\phi}^l(f_U)$ is **invertible**, the Fourier coefficients of f_X can be recovered by

$$\phi^l(f_X) = \left\{ \tilde{\phi}^l(f_U) \right\}^{-1} \phi^l(f_Z).$$

The estimator is obtained by replacing the unknown coefficients of f_Z with their empirical averages and truncating the infinite expansion at degree T_n , because the infinite sum is prone to a large variability. This leads to our deconvolution density estimator

$$\hat{f}_X(x) = \frac{1}{n} \sum_{i=1}^n \text{Re} \{ K_{T_n}(x, Z_i) \},$$

where K_{T_n} is the truncated deconvolution kernel which corrects the effect of measurement error. Similarly, motivated by the deconvolution density estimator, the proposed the regression estimator is

$$\hat{m}(x) = \hat{f}_X(x)^{-1} \frac{1}{n} \sum_{i=1}^n \operatorname{Re} \{K_{T_n}(x, Z_i)\} Y_i.$$

The key property is Proposition 3, which states that

$$E \{K_{T_n}(x, Z) \mid X\} = K_{T_n}^*(x, X).$$

Thus, although we observed contaminated observation Z , the kernel acts as we observe X in regression setting.

1.4 The main results

This work shows convergence rate and asymptotic normality based on three smoothness conditions. Also, in both density estimation and regression, it suggests asymptotic confidence intervals based on asymptotic normality and empirical likelihood methods. In addition, it shows superior performance via simulation studies and real data analysis.

1.5 My takeaways

- Why the estimator uses only real parts? doesn't miss any information loss?
- In regression estimator, why divide density estimator $\hat{f}_X$?

2 Density estimation for mixed Euclidean and non-Euclidean data in the presence of measurement error

2.1 The problem

Most existing deconvolution density estimation methods are designed for Euclidean data. However, many modern datasets contain both Euclidean and non-Euclidean variables, such as a real-valued pollutant level together with a circular wind direction. Although density estimation for mixed Euclidean and non-Euclidean data has been studied in the error-free case, the measurement error problem for such mixed data has been largely unexplored. Therefore, this study develops a deconvolution density estimator for mixed data taking values in $\mathbb{R}^d \times \mathbb{G}_c$.

2.2 The setup

Let $X = (X_1, X_2)$, where $X_1 \in \mathbb{R}^d$ is a Euclidean variable and $X_2 \in \mathbb{G}_c$ is a non-Euclidean variable. We do not observe X directly. Instead, we observe $Z = (Z_1, Z_2) = (U_1 + X_1, U_2 \circ X_2)$, where $U_1 \in \mathbb{R}^d$ and $U_2 \in \mathbb{G}_c$ are unobservable measurement errors. The goal is to estimate the joint density f_X of X with respect to the product measure $\operatorname{Leb}^d \otimes \mu$. We assume that $X \perp U$, $U_1 \perp U_2$ and that the measurement error distributions f_{U_1} and f_{U_2} are known.

2.3 The main idea

The key idea is to combine two different deconvolution techniques: the usual Fourier deconvolution on $\mathbb{R}^d$ and the group Fourier deconvolution on $\mathbb{G}_c$. For the Euclidean component, the measurement error model $Z_1 = X_1 + U_1$ gives the usual Fourier relation

$$\phi_{Z_1}(t) = \phi_{X_1}(t) \phi_{U_1}(t).$$

Therefore, the Fourier transform of X_1 can be recovered by dividing by $\phi_{U_1}(t)$. This leads to the Euclidean deconvolution kernel

$$K_h^D(s) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{-i\langle t, s \rangle} \frac{\phi_K(t)}{\phi_{U_1}(t/h)} dt.$$

For the non-Euclidean component, we use Fourier analysis on compact Lie groups. The density on $\mathbb{G}_c$ can be expanded using irreducible representations $\sigma \in \widehat{\mathbb{G}_c}$, and the convolution relation induced by $Z_2 = U_2 \circ X_2$ gives

$$\phi_{Z_2}(\sigma^M) = \phi_{X_2}(\sigma^M) \phi_{U_2}(\sigma^M).$$

Hence, when $\phi_{U_2}(\sigma^M)$ is invertible, the Fourier coefficients of X_2 can be recovered by multiplying by $\phi_{U_2}(\sigma^M)^{-1}$. Since using all representations would lead to large variability, the expansion is truncated to the representations satisfying $k_\sigma < T_n$, where k_σ is the Casimir spectrum.

Combining the Euclidean and non-Euclidean corrections, the proposed estimator is

$$\widehat{f}_X(x) = \frac{1}{nh^d} \sum_{i=1}^n K_h^D \left(\frac{x_1 - Z_{i,1}}{h} \right) \text{Re} \{ K_{T_n}(x_2, Z_{i,2}) \},$$

where $x = (x_1, x_2) \in \mathbb{R}^d \times \mathbb{G}_c$. Thus, the estimator corrects the Euclidean measurement error through K_h^D and the non-Euclidean measurement error through the truncated group deconvolution kernel K_{T_n} .

A key property of the estimator is the unbiased scoring property:

$$E \left[K_h^D \left(\frac{x_1 - Z_1}{h} \right) K_{T_n}(x_2, Z_2) \mid X \right] = K \left(\frac{x_1 - X_1}{h} \right) K_{T_n}^*(x_2, X_2).$$

This means that, although the estimator is constructed from the contaminated observation Z , the deconvolution kernels behave in conditional expectation as if the true error-free variable X were observed.

2.4 The main results

The proposed estimator derives several asymptotic properties of the estimator. In particular, we study pointwise convergence rates, partial uniform convergence rates over $\mathbb{G}_c$, full uniform convergence rates over $\mathbb{R}^d \times \mathbb{T}^q$, L^2 convergence rates over $\mathbb{R}^d \times \mathbb{T}^q$ and asymptotic normality under certain smoothness scenarios. Finally, simulation studies and a real data analysis show that the proposed estimator performs better than naive estimators that ignore measurement error.

2.5 My takeaways

- In this paper, samples are observed in $\mathbb{R}^d \times \mathbb{G}_c$, but why the target density is not the product of marginal probability?
A: Actually, a density on the product of sample space $\neq$ the product density, and this paper targets just a joint density $f_{X_1, X_2}(x_1, x_2)$!

3 Errors-in-variables regression for mixed Euclidean and non-Euclidean predictors

3.1 The problem

Most existing errors-in-variables regression methods are designed for Euclidean predictors. However, many modern datasets contain both Euclidean and non-Euclidean predictors. Although regression methods for non-Euclidean data have been studied in the absence of measurement error, the regression problem with mixed Euclidean and non-Euclidean predictors contaminated by measurement errors has been largely unexplored. Therefore, this study considers nonparametric regression with predictors taking values in $\mathbb{R}^d \times \mathbb{G}$. The goal is to estimate the regression function when both the Euclidean and non-Euclidean predictors are observed with measurement errors.

3.2 The setup

Let $X = (X_1, X_2) \in \mathbb{R}^d \times \mathbb{G}$ where $X_1 \in \mathbb{R}^d$ is a Euclidean predictor and $X_2 \in \mathbb{G}$ is a non-Euclidean predictor. The regression model is

$$Y = m(X) + \epsilon, \quad E(\epsilon | X) = 0.$$

However, the true predictor X is not directly observed. Instead, we observe

$$Z = (Z_1, Z_2) = (U_1 + X_1, U_2 \circ X_2),$$

where $U_1 \in \mathbb{R}^d$ and $U_2 \in \mathbb{G}$ are unobservable measurement errors. We assume that $U \perp (X, \epsilon)$, $U_1 \perp U_2$, and that the measurement error distributions f_{U_1} and f_{U_2} are known.

Given i.i.d. observations

$$\{(Y_i, Z_i) : 1 \leq i \leq n\}, \quad Z_i = (Z_{i,1}, Z_{i,2}) \in \mathbb{R}^d \times \mathbb{G},$$

the aim is to estimate

$$m(x) = E(Y | X = x), \quad x = (x_1, x_2) \in \mathbb{R}^d \times \mathbb{G}.$$

3.3 The main idea

The key idea is to combine Euclidean deconvolution and non-Euclidean deconvolution in a regression estimator. For the Euclidean component, the measurement error model

$$Z_1 = X_1 + U_1$$

leads to the usual Fourier deconvolution. For a kernel $K : \mathbb{R}^d \rightarrow \mathbb{R}$, the Euclidean deconvolution kernel is defined by

$$K_h^D(s) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{-i\langle s, t \rangle} \frac{\phi_K(t)}{\phi_{U_1}(t/h)} dt.$$

This kernel corrects the additive measurement error in the Euclidean predictor.

For the non-Euclidean component, we use Fourier analysis on the compact Lie group $\mathbb{G}$. For each irreducible representation $\sigma \in \widehat{\mathbb{G}}$, the group Fourier transform gives a matrix-valued convolution relation. Since the measurement error model is

$$Z_2 = U_2 \circ X_2,$$

the effect of U_2 can be corrected by multiplying by $\phi_{U_2}(\sigma^M)^{-1}$. This leads to the deconvolution kernel

$$K_{T_n}(x_2, z_2) = \sum_{\sigma \in \widehat{\mathbb{G}}: k_\sigma < T_n} d_\sigma \operatorname{Tr} \{ \sigma^M(z_2^{-1}) \phi_{U_2}(\sigma^M)^{-1} \sigma^M(x_2) \},$$

where k_σ is the Casimir spectrum and T_n is a smoothing parameter. Combining these two corrections, we propose

$$\widehat{m}(x) = \widehat{f}_X(x)^{-1} \frac{1}{nh^d} \sum_{i=1}^n K_h^D \left(\frac{x_1 - Z_{i,1}}{h} \right) \operatorname{Re} \{ K_{T_n}(x_2, Z_{i,2}) \} Y_i,$$

where

$$\widehat{f}_X(x) = \frac{1}{nh^d} \sum_{i=1}^n K_h^D \left(\frac{x_1 - Z_{i,1}}{h} \right) \operatorname{Re} \{ K_{T_n}(x_2, Z_{i,2}) \}.$$

The denominator $\widehat{f}_X(x)$ estimates the density of X , while the numerator estimates $m(x)f_X(x)$. Therefore, their ratio estimates the regression function $m(x)$.

A key principle behind the estimator is the unbiased scoring property:

$$E \left[K_h^D \left(\frac{x_1 - Z_1}{h} \right) K_{T_n}(x_2, Z_2) \mid X \right] = K \left(\frac{x_1 - X_1}{h} \right) K_{T_n}^*(x_2, X_2).$$

This means that, although the estimator is constructed from the contaminated predictor Z , the deconvolution kernels behave in conditional expectation as if the error-free predictor X were observed. This property plays an important role in correcting the bias caused by measurement error in the regression setting.

3.4 The main results

We derive pointwise, uniform, and L^2 convergence rates for the proposed estimator. We also establish the asymptotic distribution of $\hat{m}(x)$ under the ordinary-smooth scenario. The asymptotic normality result can be used to construct asymptotic confidence intervals for $m(x)$. In addition, we discuss how to handle the case where the measurement error distributions are unknown.

Simulation studies show that the proposed estimator outperforms the naive estimator that ignores measurement error.

3.5 My takeaways

- Why these papers introduce a topological group?
A: To use convolution, Fourier transform, Haar measure, deconvolution identity etc.

4 Local Fréchet regression with spherical predictors

4.1 The problem

Most existing nonparametric regression methods are designed for Euclidean responses and predictors. However, many modern datasets contain non-Euclidean objects, such as distributions, networks, covariance matrices, and manifold-valued data. Fréchet regression provides a useful framework for handling responses in a general metric space, but most existing works focus on Euclidean predictors. Therefore, this study considers Fréchet regression with spherical predictors, where the predictor takes values in $\mathbb{S}^d = \{u \in \mathbb{R}^{d+1} : \|u\|_2 = 1\}$ and the response takes values in a general metric space $(\mathbb{M}, d_{\mathbb{M}})$.

4.2 The setup

Let $Y \in \mathbb{M}$ be a response variable and $X \in \mathbb{S}^d$ be a spherical predictor. When $\mathbb{M} = \mathbb{R}$, the usual regression function can be written as

$$E(Y \mid X = x) = \arg \min_{y \in \mathbb{R}} E\{(Y - y)^2 \mid X = x\}.$$

To extend this idea to a general metric space, this paper defines

$$M_{\oplus}(x, y) = E\{d_{\mathbb{M}}^2(Y, y) \mid X = x\},$$

and the target regression function is

$$m_{\oplus}(x) = \arg \min_{y \in \mathbb{M}} M_{\oplus}(x, y).$$

Given i.i.d. observations

$$\{(X_i, Y_i) : 1 \leq i \leq n\},$$

the goal is to estimate $m_{\oplus}(x)$ for $x \in \mathbb{S}^d$.

4.3 The main idea

The main idea is to construct local Fréchet regression estimators using the geometry of the sphere. For the local constant estimator, nearby observations are selected according to the spherical closeness between X_i and x . Since the angle between X_i and x is $\arccos(X_i^\top x)$, the quantity

$$1 - X_i^\top x$$

is small when X_i is close to x . Using a kernel L and a bandwidth h , the local constant objective function is

$$\widehat{M}_{h,0}(x, y) = \frac{n^{-1} \sum_{i=1}^n L\left(\frac{1 - X_i^\top x}{h^2}\right) d_{\mathbb{M}}^2(Y_i, y)}{n^{-1} \sum_{i=1}^n L\left(\frac{1 - X_i^\top x}{h^2}\right)}.$$

The local constant Fréchet estimator is then defined by

$$\widehat{m}_{h,0}(x) = \arg \min_{y \in \mathbb{M}} \widehat{M}_{h,0}(x, y).$$

We also propose a local linear Fréchet estimator. For this, it uses a tangent-normal decomposition around $x \in \mathbb{S}^d$. Each nearby point $z \in \mathbb{S}^d$ can be written approximately through its tangent direction at x , and a first-order approximation motivates local linear weights. The resulting local linear objective function is

$$\widehat{M}_{h,1}(x, y) = n^{-1} \sum_{i=1}^n \widehat{W}_{x,h}(X_i) d_{\mathbb{M}}^2(Y_i, y),$$

where $\widehat{W}_{x,h}(X_i)$ is the spherical local linear weight. The local linear Fréchet estimator is

$$\widehat{m}_{h,1}(x) = \arg \min_{y \in \mathbb{M}} \widehat{M}_{h,1}(x, y).$$

Thus, both estimators are defined as minimizers of weighted Fréchet objective functions, but the local linear estimator uses tangent-space information to improve local approximation.

4.4 The main results

We establish the consistency and convergence rates of both $\widehat{m}_{h,0}(x)$ and $\widehat{m}_{h,1}(x)$. We also provide simulation studies for spherical-circular and spherical-spherical regression, where the proposed local linear estimator performs best. In real data analyses, the method is applied to distributional-circular regression, distributional-spherical regression, and network-circular regression.

4.5 My takeaways

- Why this paper use $1 - X_i^\top x$ instead of the geodesic distance $\arccos(X_i^\top x)$? Is it for computational efficiency or theoretical simplicity?

5 Additive regression for non-Euclidean responses and predictors

5.1 The problem

Most existing nonparametric regression methods are designed for Euclidean responses and predictors. However, modern datasets often contain non-Euclidean objects, such as functional data, density-valued data, compositional data, spherical data, toroidal data, and shape data. Although nonparametric regression for some non-Euclidean variables has been studied, most existing methods are either full-dimensional or restricted to special types of responses and predictors. Therefore, this study considers additive regression in a very general setting where the response takes values in a separable Hilbert space $\mathbb{H}$, and the predictors take values in general semimetric spaces.

5.2 The setup

Let $Y \in \mathbb{H}$ be a Hilbertian response variable and let

$$X = (X_1, \dots, X_d) \in \mathcal{X} := \prod_{j=1}^d \mathcal{X}_j,$$

where each X_j takes values in a semimetric space $(\mathcal{X}_j, d_j)$. We consider the additive model

$$Y = E(Y) \oplus \sum_{j=1}^d f_j(X_j) \oplus \epsilon,$$

where $f_j : \mathcal{X}_j \rightarrow \mathbb{H}$ are unknown component maps and ϵ is an $\mathbb{H}$ -valued error term satisfying

$$E(\epsilon \mid X) = 0.$$

For identifiability, the component maps satisfy

$$E\{f_j(X_j)\} = 0, \quad 1 \leq j \leq d.$$

The goal is to estimate the additive regression map

$$f(x) = E(Y) \oplus \sum_{j=1}^d f_j(x_j),$$

as well as the individual component maps f_j .

5.3 The main idea

The key idea is to extend smooth backfitting to a setting where the response is Hilbert-space-valued and the predictors are semimetric-space-valued. Let

$$m_j(x_j) = E(Y \mid X_j = x_j)$$

be the marginal regression map. By taking conditional expectation of the additive model given $X_j = x_j$, the component maps satisfy the system of Bochner integral equations

$$f_j(x_j) = m_j(x_j) \ominus E(Y) \ominus \sum_{k \neq j} \int_{\mathcal{X}_k} f_k(x_k) \frac{p_{jk}(x_j, x_k)}{p_j(x_j)} d\mu_k(x_k), \quad 1 \leq j \leq d,$$

where p_j and p_{jk} are the marginal and joint densities of X_j and (X_j, X_k) , respectively. Thus, the additive components can be recovered by solving a system of integral equations.

We estimate the unknown quantities in this system using kernel-type estimators on the predictor spaces. For each predictor space $\mathcal{X}_j$, a kernel weight function $L_{j,n}$ is used to estimate the joint density p , the marginal densities p_j , and the marginal regression maps m_j . The resulting estimated system is

$$\hat{f}_j(x_j) = \hat{m}_j(x_j) \ominus \bar{Y} \ominus \sum_{k \neq j} \int_{\mathcal{X}_k} \hat{f}_k(x_k) \frac{\hat{p}_{jk}(x_j, x_k)}{\hat{p}_j(x_j)} d\mu_k(x_k), \quad 1 \leq j \leq d.$$

The solution $(\hat{f}_1, \dots, \hat{f}_d)$ is called the semimetric Bochner smooth backfitting estimator.

Since the estimator has no closed form, we propose an iterative smooth backfitting algorithm. Starting from initial estimators $\hat{f}_j^{[0]}$, the r -th update is

$$\hat{f}_j^{[r]}(x_j) = \hat{m}_j(x_j) \ominus \bar{Y} \ominus \sum_{k < j} \int_{\mathcal{X}_k} \hat{f}_k^{[r]}(x_k) \frac{\hat{p}_{jk}(x_j, x_k)}{\hat{p}_j(x_j)} d\mu_k(x_k) \ominus \sum_{k > j} \int_{\mathcal{X}_k} \hat{f}_k^{[r-1]}(x_k) \frac{\hat{p}_{jk}(x_j, x_k)}{\hat{p}_j(x_j)} d\mu_k(x_k).$$

5.4 The main results

We establish the existence and uniqueness of the semimetric Bochner smooth backfitting estimator. It also proves that the proposed backfitting algorithm converges to the estimator. For Hilbertian and Riemannian predictors, we derive rates of convergence for both the full regression estimator

$$\hat{f} = \bar{Y} \oplus \sum_{j=1}^d \hat{f}_j$$

and the individual component estimators $\hat{f}_j$. Especially, for Hölder continuous component maps, the convergence rates depend on the individual dimensions q_j of the predictor spaces, rather than on the total dimension $\sum_{j=1}^d q_j$. This shows that the proposed additive structure avoids the curse of dimensionality.

We also provide a more general convergence theorem for predictors taking values in arbitrary semimetric spaces. Simulation studies and real data analyses demonstrate that the method can handle mixed Euclidean and non-Euclidean predictors and Hilbert-space-valued responses.

5.5 My takeaways

Further thinking is required.